

Assignment 4 Code Review

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1. The original **loadTopScores()** method was too long and messy.

Relevant commit - Refactored **loadTopScores()**

Solution : Extracted the file reading logic into a separate helper method, updated it to read lines instead of chars. All of the above increases readability and opens the potential to reuse the file reading logic elsewhere.

Refactoring performed : Created separate **readFileContent()** method and passed file reading logic there. Updated the **loadTopScores()** to use the introduced helper method.

2. JSON parsing was still inside **loadTopScores()**.

Relevant commit - Helper for JSON parsing

Solution : Extracted the file parsing logic into a separate helper method. Increases readability, maintainability.

Refactoring performed : Created separate **parseToJson()** method and passed file reading logic there. Updated the **loadTopScores()** to use the introduced helper method.

3. Hardcoded value in **getTopScores()**, which is a bad practice for maintainability and readability.

Relevant commit - Removed hardcoded string value

Solution : Created separate **final** variable to handle the default value case.

Refactoring performed : Removed the hardcoded "000" string. Created separate **DEFAULT_SCORE** to handle the default value.

4. Code smell - Feature Envy. **updateTable()** interacted with table data directly very often. Moreover in made redundant calls that were unnecessary.

Relevant commit - Refactored updateTable()

Solution : Created method that would interact with the table directly and reduced updateTable() interactions. This centralized the data update logic. Remove redundant calls made.

Refactoring performed : Created separate method to interact with table values directly. Updated **updateTable()** to use this method. Removed redundant calls that were handled when calling **setValueAt()**.

5. Long method **styleTable()**

Relevant commit - Refactored styleTable()

Solution : Since the method was very hard to read and change, we changed it to become modular and easy to get. This improves clarity and makes the styling easy to change, since now you need to interact with only 1 helper method. This improved clarity and made maintenance easier.

Refactoring performed : Created multiple helper methods in order to reduce the size of main **styleTable()** method. Moved the appropriate functionality to those methods. Updated the **styleTable()** method to use introduced methods.

6. Overlapping responsibilities in **loadTopScores()** and not use existing functions
Relevant commit - Refactored loadTopScores() with new methods

Solution : After implementing methods for previous steps, never used them here. Make use of existing methods.

Refactoring performed : Created separate method for processing the data and loading it. Separated responsibilities, which is a good way of writing code using OOP principles.

7. Poor error handling
Relevant commit - Error handling update

Solution : Need to handle errors more accurately during runtime and give users feedback regarding their actions if something goes wrong.

Refactoring performed : Created method **handleError()** which now handles errors and prints appropriate message. Created isTestMode variable which allows to enable this feature during test phase, and enable it when app is launched.

8. No reset scores functionality - missing functionality!

Relevant commit - Error handling update

Solution : Since different users can play game on one computer, it requires to have a function of resetting the best scores. Create a method and a button to use this method.

Refactoring performed : Created **resetScore()** method that clears the best scores in the table. Created a button that handles the functionality of the above method.